Data Science: Visualization

Gustavo Roa

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Section 1

Describe Heights to ET

```
# load the dataset
library(dslabs)
data(heights)

# make a table of category proportions
prop.table(table(heights$sex))
```

```
##
##    Female      Male
## 0.2266667 0.7733333
```

```
my_data <- data(heights)
```

Cumulative Distribution Function

```
#a <- seq(min(my_data), max(my_data), length = 100)    # define range of values spanning the dataset
#cdf_function <- function(x) {    # computes prob. for a single value
    #mean(my_data <= x)
#}
#cdf_values <- sapply(a, cdf_function)
#plot(a, cdf_values)
```

Normal distribution

```
# define x as vector of male heights
library(tidyverse)
```

```
## Warning: package 'ggplot2' was built under R version 4.2.3
```

```
## Warning: package 'tibble' was built under R version 4.2.3
```

```
## Warning: package 'dplyr' was built under R version 4.2.3

## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.2     v readr     2.1.4
## v forcats   1.0.0     v stringr   1.5.0
## v ggplot2   3.4.2     v tibble    3.2.1
## v lubridate 1.9.2     v tidyr     1.3.0
## v purrr     1.0.1
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
library(dslabs)
data(heights)
index <- heights$sex=="Male"
x <- heights$height[index]

# calculate the mean and standard deviation manually
average <- sum(x)/length(x)
SD <- sqrt(sum((x - average)^2)/length(x))

# built-in mean and sd functions - note that the audio and printed values disagree
average <- mean(x)
SD <- sd(x)
c(average = average, SD = SD)
```

```
##   average        SD
## 69.314755  3.611024
```

```r
# calculate standard units
z <- scale(x)
#  2 standard deviations
mean(abs(z) < 2)
```

```
## [1] 0.9495074
```

```r
# calculate proportion of values within 2 SD of mean
mean(abs(z) < 2)
```

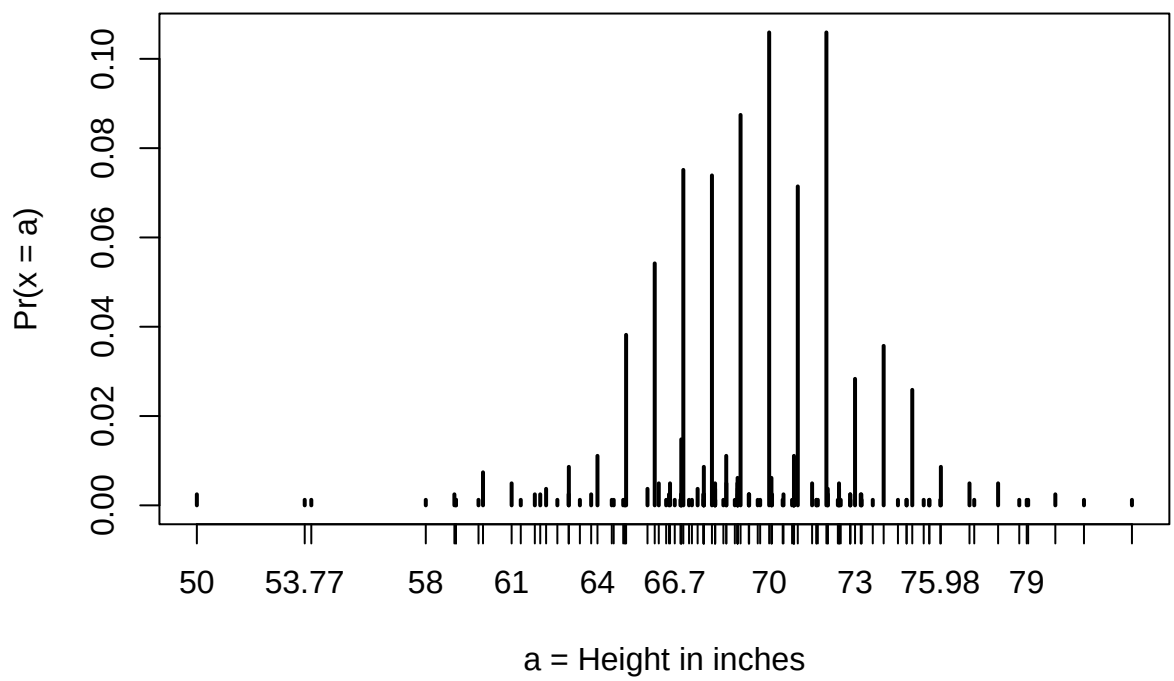
```
## [1] 0.9495074
```

The Normal CDF and pnorm

```r
library(tidyverse)
library(dslabs)
data(heights)
x <- heights %>% filter(sex=="Male") %>% pull(height)
#We can estimate the probability that a male is taller than 70.5 inches with:

1 - pnorm(70.5, mean(x), sd(x))
```

```
## [1] 0.371369
```

```r
#Code: Discretization and the normal approximation
# plot distribution of exact heights in data
plot(prop.table(table(x)), xlab = "a = Height in inches", ylab = "Pr(x = a)")
```




```r
# probabilities in actual data over length 1 ranges containing an integer
mean(x <= 68.5) - mean(x <= 67.5)
```

```
## [1] 0.114532
```

```r
mean(x <= 69.5) - mean(x <= 68.5)
```

```
## [1] 0.1194581
```

```r
mean(x <= 70.5) - mean(x <= 69.5)
```

```
## [1] 0.1219212
```

```r
# probabilities in normal approximation match well
pnorm(68.5, mean(x), sd(x)) - pnorm(67.5, mean(x), sd(x))
```

```
## [1] 0.1031077
```

```r
pnorm(69.5, mean(x), sd(x)) - pnorm(68.5, mean(x), sd(x))
```

```
## [1] 0.1097121
```

```r
pnorm(70.5, mean(x), sd(x)) - pnorm(69.5, mean(x), sd(x))
```

```
## [1] 0.1081743
```

```r
# probabilities in actual data over other ranges don't match normal approx as well
mean(x <= 70.9) - mean(x <= 70.1)
```

```
## [1] 0.02216749
```

```r
pnorm(70.9, mean(x), sd(x)) - pnorm(70.1, mean(x), sd(x))
```

```
## [1] 0.08359562
```

Definition of quantiles

```r
#Examples
#Load the heights dataset from the dslabs package:

library(dslabs)
data(heights)
#Use summary() on the heights$height variable to find the quartiles:

summary(heights$height)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   50.00   66.00   68.50   68.32   71.00   82.68
```

```r
#Find the percentiles of heights$height:

p <- seq(0.01, 0.99, 0.01)
percentiles <- quantile(heights$height, p)
#Confirm that the 25th and 75th percentiles match the 1st and 3rd quartiles. Note that quantile() retur

percentiles[names(percentiles) == "25%"]
```

```
## 25%
##  66
```

```r
percentiles[names(percentiles) == "75%"]
```

```
## 75%
##  71
```

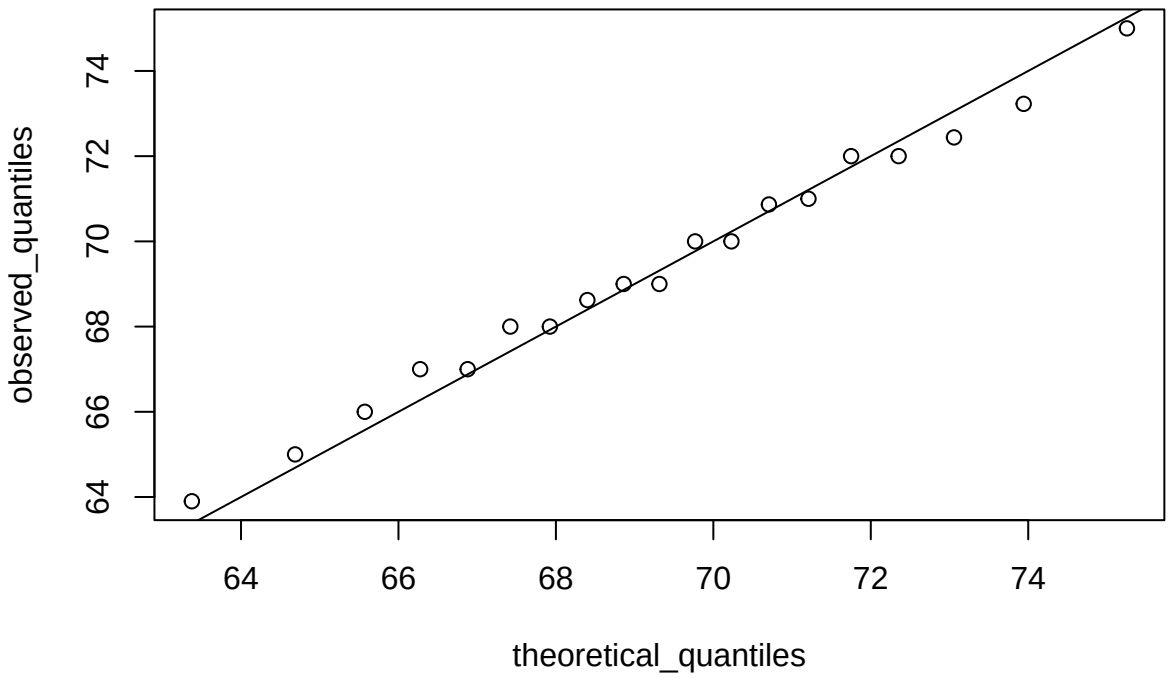
```r
#Theoretical quantiles
p <- seq(0.01, 0.99, 0.01)
theoretical_quantiles <- qnorm(p, 69, 3)
```

Quantile-Quantile Plots

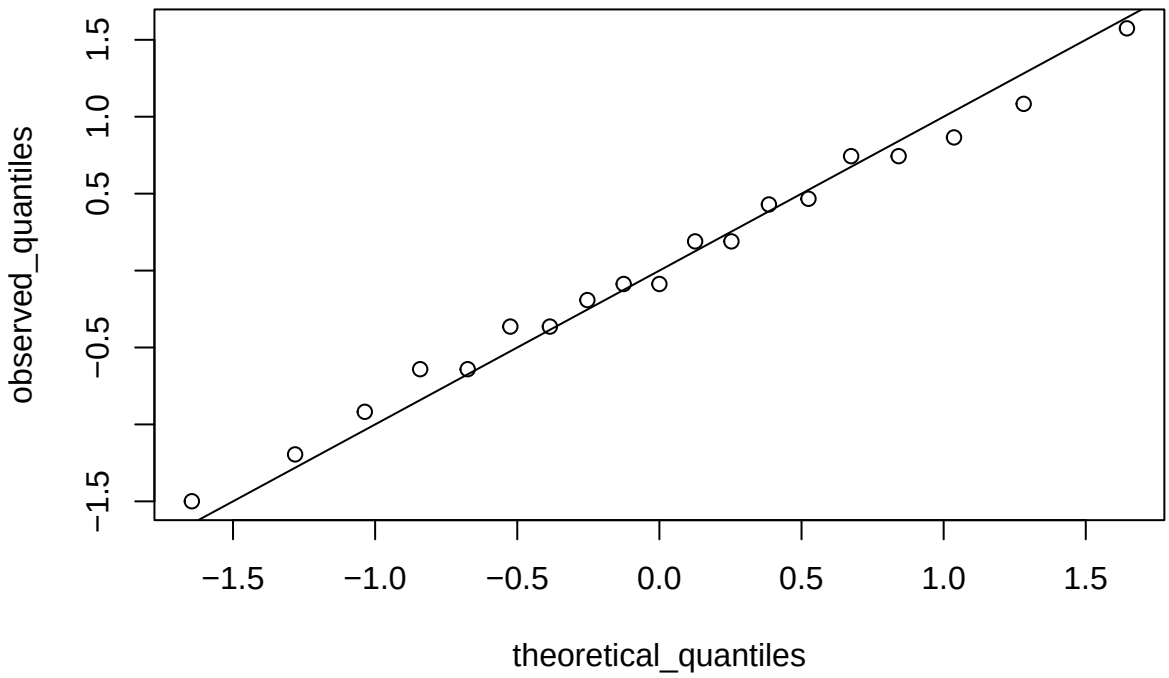
```r
# define x and z
library(tidyverse)
library(dslabs)
data(heights)
index <- heights$sex=="Male"
x <- heights$height[index]
z <- scale(x)

# proportion of data below 69.5
mean(x <= 69.5)
```

```
## [1] 0.5147783
```

```r
# calculate observed and theoretical quantiles
p <- seq(0.05, 0.95, 0.05)
observed_quantiles <- quantile(x, p)
theoretical_quantiles <- qnorm(p, mean = mean(x), sd = sd(x))

# make QQ-plot
plot(theoretical_quantiles, observed_quantiles)
abline(0,1)
```




```r
# make QQ-plot with scaled values
observed_quantiles <- quantile(z, p)
theoretical_quantiles <- qnorm(p)
plot(theoretical_quantiles, observed_quantiles)
abline(0,1)
```



Section 2 ## Creating a New Plot

```
library(tidyverse)
library(dslabs)
data(murders)

ggplot(data = murders)
```

```
murders %>% ggplot()
```

```r
p <- ggplot(data = murders)
class(p)
```

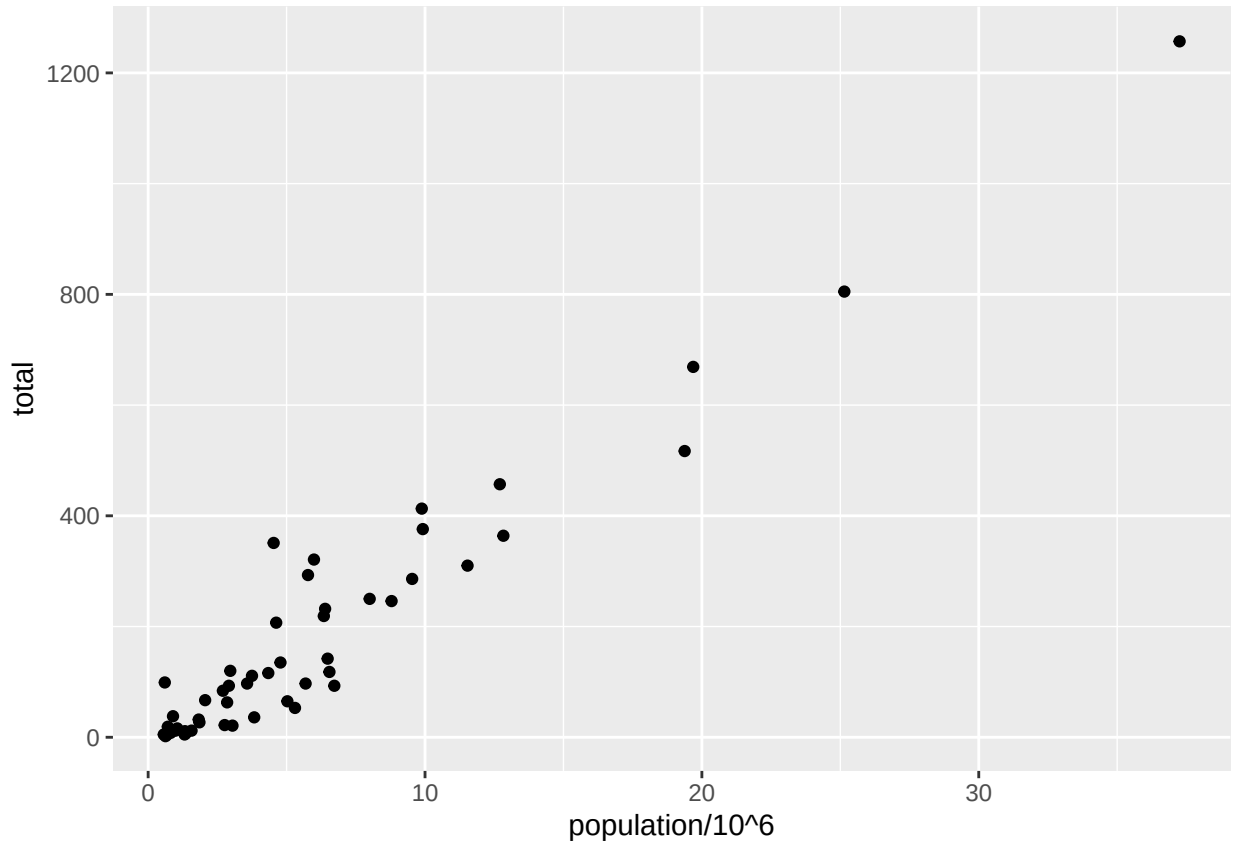
```
## [1] "gg"     "ggplot"
```

```r
print(p)    # this is equivalent to simply typing p
```

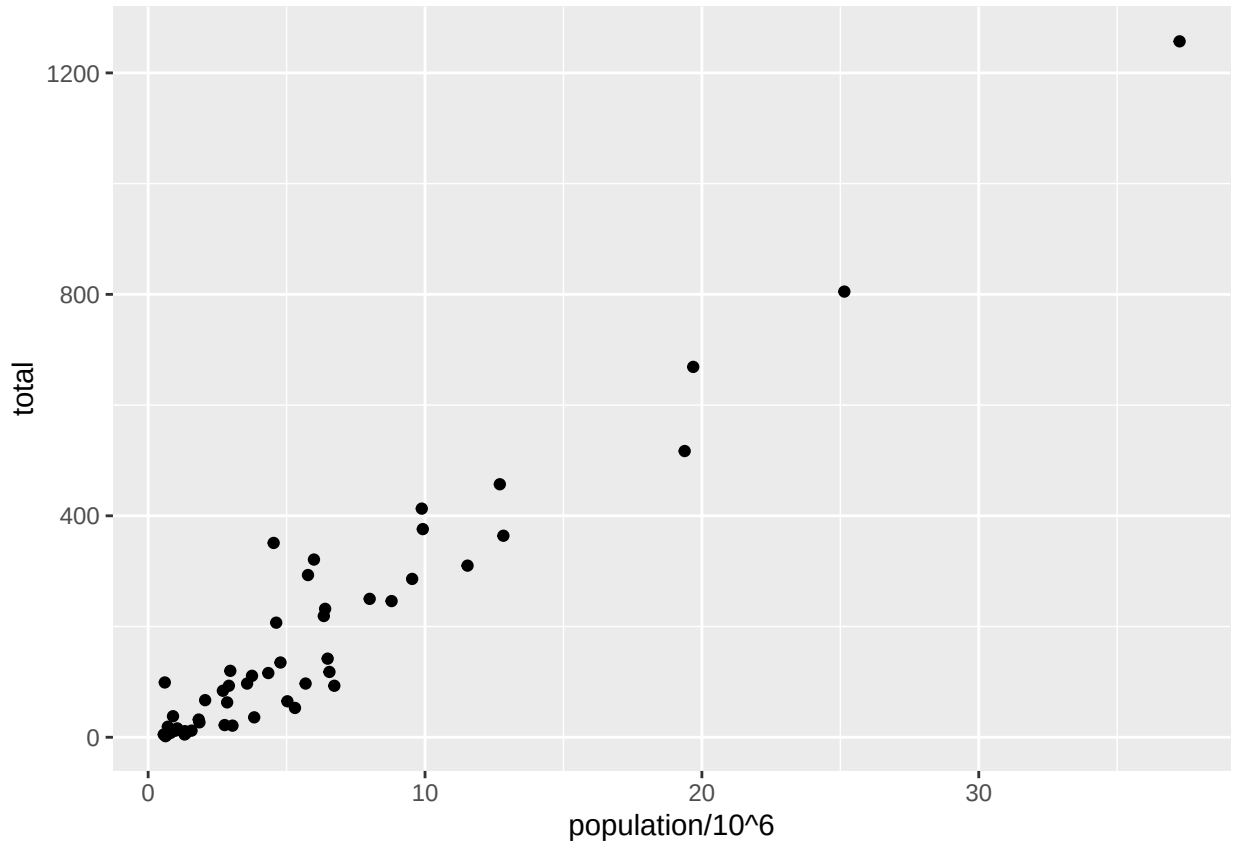
```
p
```

Layers

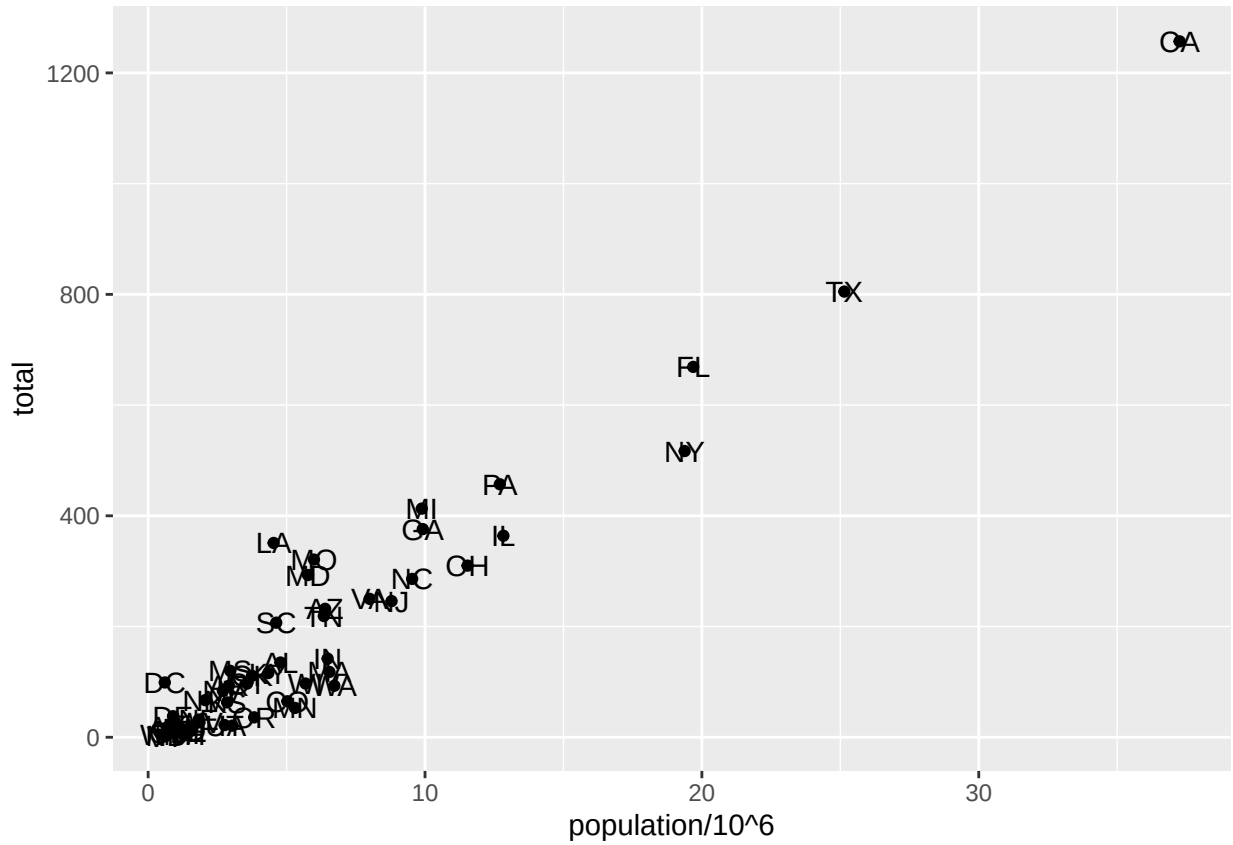
```
#Code: Adding layers to a plot
library(tidyverse)
library(dslabs)
data(murders)

murders %>% ggplot() +
    geom_point(aes(x = population/10^6, y = total))
```




```
# add points layer to predefined ggplot object
p <- ggplot(data = murders)
p + geom_point(aes(population/10^6, total))
```




```
# add text layer to scatterplot
p + geom_point(aes(population/10^6, total)) +
    geom_text(aes(population/10^6, total, label = abb))
```




```
#Code: Example of aes behavior
# no error from this call
p_test <- p + geom_text(aes(population/10^6, total, label = abb))

# error - "abb" is not a globally defined variable and cannot be found outside of aes
#p_test <- p + geom_text(aes(population/10^6, total), label = abb)
```

Tinkering